ChatGpt: Decreasing Universe Theory (DUT) — Comprehensive Summary

1. Central idea

The **Decreasing Universe Theory (DUT)** proposes that the Universe is **not actually expanding**. Instead, what changes over time are the **physical scales of matter** — particles, atoms, and characteristic lengths — which would **gradually decrease**.

Since all our measuring devices (rulers, clocks, spectroscopic standards) are also made of the same matter, the shrinking of physical scales would make the Universe **appear to expand** when comparing observations from different epochs.

Conceptually:

Universe appears to expand

Thus, observations usually interpreted as cosmic expansion would instead be interpreted as an **effect of changing physical scales**.

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2. Motivation of the theory

The DUT emerges as a conceptual alternative to cosmological models based on the **metric expansion of space**, such as the standard **ΛCDM model**.

Its motivations include:

- Searching for alternative explanations for the **cosmological redshift**.
- Avoiding the need for certain elements of the standard model, such as specific interpretations of **dark energy**.
- Exploring the possibility that **fundamental physical scales or constants evolve with cosmic time**.

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3. Explanation of redshift

In the standard cosmological model (ΛCDM), redshift is interpreted mainly as a consequence of the **expansion of space** between the emission and observation of light.

In the DUT, redshift would instead arise from changes in physical scales:

1. **Atoms in the past would be larger**, or would emit photons with different intrinsic properties.
2. Over time, the scale of matter would decrease.

3. Light emitted in the past would therefore be observed today with a **longer wavelength**, producing redshift.

Thus:

without requiring expansion of space itself.

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4. Theoretical implications

If the DUT were correct, several interpretations in cosmology would need to be reconsidered. Proposed implications include:

- **Cosmological redshift** reinterpreted as a scale-evolution effect.
- A possible alternative explanation for phenomena currently attributed to **dark energy**.
- A new interpretation of the relationship between **stellar age and observed redshift**.
- The possibility that some effects attributed to **dark matter** could instead be observational artifacts related to scale evolution or photon propagation effects.

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5. Comparison with the standard cosmological model (ΛCDM)

Aspect	**ΛCDM (standard model)**	**DUT**
Expansion of the Universe	Space expands	Physical scales of matter shrink
Redshift	Metric expansion of space	Evolution of physical scales or emission properties
Dark energy	Required to explain accelerated expansion	May not be necessary
Interpretation of cosmic evolution	Universe becomes larger over time	Physical scales become smaller over time

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6. Possible observational tests

For the DUT to be considered a viable alternative, it must reproduce the large body of existing cosmological observations. Important tests include:

- **Type Ia supernovae** (distance–redshift relations)
- **Cosmic Microwave Background (CMB)**
- **Baryon Acoustic Oscillations (BAO)**
- **Large-scale structure formation**
- **Relationships between galaxy properties and redshift**

Additional tests may involve comparing galaxies with:

- **the same mass**
- **the same distance measured independently of redshift**
- **different stellar ages**

to investigate correlations between **stellar population age and residual redshift**.

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7. Scientific status

Currently:

- The DUT is considered a **speculative or alternative cosmological hypothesis**.
- It has not yet gained broad acceptance in the cosmological community.
- To compete with the ΛCDM model, it would need to demonstrate that it can simultaneously explain the wide range of observational data supporting the standard cosmological framework.

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8. Conceptual significance

Even if alternative theories such as the DUT do not replace the standard model, they can still be valuable because they:

- challenge fundamental assumptions in cosmology,
- stimulate new observational tests,
- explore the possibility that **fundamental constants or physical scales may evolve with time**.

✅ **Summary:**
The Decreasing Universe Theory proposes that the Universe may appear to expand because the physical scales of matter gradually decrease over time. This hypothesis

offers an alternative interpretation of cosmological redshift and certain cosmological phenomena, but it still needs to demonstrate consistency with the broad set of observations that currently support the standard cosmological model.

See also:

Some Articles about D.U.T:
https://jocaxx.github.io/DUniverse/